ManaDwar — Build Log (running changelog)

Living record of what was built, version by version. Updated with every new version. Pairs with "Complete Project Context" (the vision/spec); this file is what actually exists in the code.

Current version: v7.2 (git tag `v7.2`) **Stack as built:** FastAPI + SQLAlchemy (SQLite for the pilot; Postgres + Alembic planned) + Pydantic v2, JWT auth (bcrypt), single-page vanilla-JS frontend served at `/app/`. **Pilot:** software-only pilot at one apartment; RFID hardware not chosen yet (reader stays behind the `/verify-rfid` API). **Working agreement:** ask the user before making changes (they have other changes to include); after approval, test, commit + tag, update this log, send the zip.

Roles

	Admin	Guard	Resident
Gate verification (RFID / OTP / staff ID)	yes	yes (OTP by QR scan or typing)	no
Deliveries	announce for anyone, overview, override	log unexpected, let in / refuse	announce own, approve/deny own unexpected, cancel own
Visitors	all	register for anyone	own flat only; WhatsApp button + QR pass after registering
Complaints	view all, update status, raise for anyone	no	raise own (with photo), view own
Staff directory	register, block/unblock, search	read-only, search	no

Demo logins (seed): admin@manadwar.test/admin123, guard@manadwar.test/guard123, arun@manadwar.test and priya@manadwar.test /resident123.

Version history

Tag	Change
v1.0–v4.1	RFID vehicle access, OTP visitors, deliveries (old model), maintenance, entry logs (v1.0–v4.1.1 in git were rebuilt from the consolidated code, not the user's originals)
v5.0	General staff (maid/driver/security) with hashed Aadhaar, gate ID-card credential, block/unblock
v6.0	JWT login, roles (admin/guard/resident), resident portal, bcrypt
v6.1	Visitor OTP made entry-only and reusable (no exit OTP, exits not tracked); host search by name/flat; bills hidden from screens; complaint categories; staff search
v6.2	Complaint photos (JPG/PNG/WebP, max 5 MB, `uploads/` folder, owner/admin-only access); startup schema check
v6.2.1	Fix: long-term visitor registration returned 500 when the browser sent timezone-aware datetimes
v7.0	Delivery redesign (below)
v7.0.1	Phone fix: wide tables scroll sideways inside their card (Let in / Refuse buttons were cut off at 390px width). No database change.
v7.1	Resident portal: after registering a visitor, a **Send on WhatsApp** button and a **QR visitor pass**. Frontend only.
v7.2	Guard QR scanner: live camera scan + take-photo fallback; auto ACCESS GRANTED / DENIED; typed OTP kept as failsafe. Frontend only.

v7.2 Guard QR scanner (Visitor > Verify OTP)

- **Scan QR code**: live camera (rear camera preferred), decoded in the browser with vendored jsQR 1.4.0 (`app/static/jsqr-lib.js`, Apache-2.0, licence in `jsqr-LICENSE.txt`). Frames are never uploaded.
- **Take photo of QR**: file input with `capture`; works on phones over plain http (live camera does not).
- **Browser rule:** live camera only works on `localhost` or HTTPS. On a phone over `http://<laptop-ip>:8000` the page explains this and points to the photo button / typing. Live scanning on phones would need HTTPS (tunnel or reverse proxy with a certificate) — not set up.
- A valid pass → same `/verify-otp` check and entry log as typing: ACCESS GRANTED with visitor + host name, otherwise ACCESS DENIED. A QR that isn't a ManaDwar pass (or bare 6-digit code) shows "Not a ManaDwar visitor pass" and is not logged (no OTP to log).
- Cooldown: the same pass held in view counts once (5 s). Photo scans have no cooldown (explicit action). Camera stops on tab change, Stop button, or when the page is hidden.

- Typed OTP stays as the failsafe (heading "Or type the OTP"). Resident's QR note now says the guard can scan it.
- Accepted QR text: "ManaDwar Visitor Pass ... OTP: 123456 ..." (what v7.1 shows) or a bare 6-digit code. QRs from v7.1 still work.
- Admin sees the same scanner (shared console). Guard registration screen still has no QR/WhatsApp (user chose resident only).

v7.1 WhatsApp invite + QR pass (resident portal only)

- Button opens `https://wa.me/<visitor number>?text=...` (new tab). Nothing is sent automatically; the resident presses send in WhatsApp.
- Message: "Hello {visitor}, / {resident} (Flat {flat}) is inviting you to visit. / This is your ManaDwar entry OTP: {otp} / Show it to the guard at the gate. It works for every entry until {time}." Wording lives in `inviteMessage()` in `app/static/index.html` (user chose own wording but did not type it; built from their first description, change on request).
- Phone normalisation: 10 digits → +91 added; `0`-prefixed 11 digits; `+91 …` / `91…`; other countries only when typed with `+` or `00`. Unusable numbers show a note instead of a button (QR still shown).
- QR = visitor pass text (visitor, host + flat, OTP, valid until), generated in the browser (no data leaves it) with vendored MIT library `app/static/qrcode-lib.js` (qrcode-generator 1.4.4 + UTF-8 support). Drawn at 5 px per square for easy scanning.
- WhatsApp links carry text only, so the QR cannot be attached to the message; resident shows or screenshots it.
- Not built (possible later): re-send from the My Visitors list, automatic sending through an SMS/WhatsApp API (needs provider account; India SMS needs DLT registration).

Mobile compatibility (checked at 390px, v7.0.1 – v7.2)

- Responsive web page (viewport tag, single-column forms, no page-level sideways scroll) for resident, guard and admin views.
- Tables scroll horizontally within their card on small screens.
- It is a web page, not an installed app; runs in the phone browser when the phone can reach the server (same Wi-Fi, e.g. `uvicorn app.main:app --host 0.0.0.0` then `http://<laptop-ip>:8000`).

v7.0 delivery flow (replaces the old auto/manual approval model)

1. Resident **announces** a delivery: platform, note, time window (`POST /deliveries/announce`) → `EXPECTED`.
2. Guard sees the "waiting at the gate" list (searchable) and taps **Let in** (`/admit` → `ADMITTED`, logged as DELIVERY/GRANTED) or **Refuse** (`/refuse` → `DENIED`, logged).
3. Courier nobody announced: guard logs an **unexpected arrival** (`POST /deliveries/walk-in` → `AWAITING_APPROVAL`). Resident approves/denies (`/decision`). Guard can admit only after approval.
4. Resident or admin can **cancel** open deliveries (`/cancel`).

5. Lazy expiry: announced deliveries expire at window end; unanswered walk-ins after 2 hours.
6. Admin overview: `GET /deliveries/summary` tiles + searchable history, with approve/deny/cancel overrides.

- Removed: old `POST /deliveries` with `approval_mode` (intentional breaking change).
- Statuses: EXPECTED, AWAITING_APPROVAL, APPROVED, ADMITTED, DENIED, CANCELLED, EXPIRED.

Other behaviour worth knowing

- Bills: UI removed; backend endpoints and data kept.
- Complaint categories: Parking, Cleaning, Lift, Garbage, Water, Electricity, Noise, Security, Other.
- No migrations tool yet: schema changes mean deleting `manadwar.db` and re-running `python -m app.seed` (app checks the schema at startup and says so).

Verification (v7.2)

Static checks plus four suites, each on a fresh DB and server: roles 46, regression 53, deliveries 60, browser end-to-end 125. v7.2 adds: photo scans (valid, wrong OTP, foreign QR, bare code, no QR, non-image), a live-camera scan with a fake webcam showing a real visitor QR (granted once while held in view), camera releases on tab change/Stop, and the no-camera-API (plain http) message. All passed, also from a fresh unzip with a new virtualenv.

Run on Windows

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
copy .env.example .env      (set JWT_SECRET_KEY)
python -m app.seed
uvicorn app.main:app --reload     -> http://127.0.0.1:8000/
```

GitHub / git status

- Local repo in the delivered zip has tags v1.0–v7.2.
- User's GitHub repo (watchermstb-maker/Manadwar) was created from a fresh `git init`, so it lacks the v1.0–v5.0 tags.
- Claude's session cannot push to that repo (GitHub app not installed for it). Pushes are done from the user's laptop: commit, tag, `git push`, `git push origin <tag>`.

Under discussion

- Automatic sending of the OTP to the visitor (SMS gateway or WhatsApp Business API). Deferred: the WhatsApp button covers the pilot. Options: MSG91/Fast2SMS/Twilio (India needs DLT registration), WhatsApp Business API, email.
- HTTPS for live QR scanning on guards' phones (not started).

Next (not started)

- Postgres + Alembic migrations
- Real RFID reader/ESP32 integration once hardware is tested
- Mobile app (Flutter) and React admin dashboard per the original plan